

CO3-AT3-Assingment – 02

1. Online Login system.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         try {
9             System.out.print("Enter username: ");
10            String username = sc.next();
11
12            System.out.print("Enter password: ");
13            String password = sc.next();
14
15            System.out.print("Enter user ID (0 or 1): ");
16            int id = sc.nextInt();
17
18            String[] users = {"admin", "user"};
19            String check = users[id];
20
21            if (username.equals("admin") && password.equals("1234")) {
22                System.out.println("Login successful!");
23            } else {
24                System.out.println("Invalid username or password.");
25            }
26        }
27    }
28 }
```

Input

```
admin
1234
0
```

Output

```
Enter username: Enter password:
Enter user ID (0 or 1): Login
successful!
Finally block executed.
```

2. Thread Priority.

Multiple Threads and Exceptions
25 marks

4/4 answered

SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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```
1 public class Main {
2
3     static class HospitalTask extends Thread {
4         String task;
5
6         HospitalTask(String task) {
7             this.task = task;
8         }
9
10        public void run() {
11            System.out.println(task + " - Priority: " + getPriority());
12        }
13    }
14
15    public static void main(String[] args) {
16
17        HospitalTask emergency = new HospitalTask("Emergency Task");
18        HospitalTask regular = new HospitalTask("Regular Task");
19        HospitalTask admin = new HospitalTask("Administrative Task");
20
21        emergency.setPriority(Thread.MAX_PRIORITY); // 10
22        regular.setPriority(Thread.NORM_PRIORITY); // 5
23        admin.setPriority(Thread.MIN_PRIORITY); // 1
24
25        System.out.println("Emergency Priority: " + emergency.getPriority());
26        System.out.println("Regular Priority: " + regular.getPriority());
27        System.out.println("Administrative Priority: " + admin.getPriority());
28    }
29 }
```

stdin input

Output

```
Running...
```

3. Traffic Signal Simulation

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1094". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "MONESH S 192324027RA". The left sidebar shows a list of activities, with "ATM Transaction System" selected, indicating 25 marks and 4/4 answered. The main area displays a Java program for a Traffic Signal Simulation. The code defines a `Signal` class that extends `Thread` and a `Main` class that creates and starts three `Signal` objects (red, yellow, green). The `Signal` class has a `run` method that prints the signal color and status, and a `sleep` method to simulate the signal duration.

```
1 public class Main {
2
3     static class Signal extends Thread {
4         String color;
5
6         Signal(String color) {
7             this.color = color;
8         }
9
10        public void run() {
11            try {
12                System.out.println(color + " Signal ON");
13                Thread.sleep(100);
14
15                System.out.println(color + " Signal OFF");
16
17            } catch (InterruptedException e) {
18                System.out.println("Thread interrupted");
19            }
20        }
21    }
22
23    public static void main(String[] args) {
24
25        try {
26            Signal red = new Signal("RED");
27            Signal yellow = new Signal("YELLOW");
28            Signal green = new Signal("GREEN");
29        }
```

4. ATM Transaction System.

The screenshot shows the SIMATS Online Programming Lab interface. The browser address bar displays "simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=1094". The page header includes the SIMATS ENGINEERING logo, the title "SIMATS Online Programming Lab", and the user profile "MONESH S 192324027RA". The left sidebar shows a list of activities, with "ATM Transaction System" selected, indicating 25 marks and 4/4 answered. The main area displays a Java program for an ATM Transaction System. The code defines an `InsufficientBalanceException` class that extends `Exception`, and an `Account` class that manages a balance and provides `deposit` and `withdraw` methods. The `withdraw` method throws an `InsufficientBalanceException` if the withdrawal amount exceeds the current balance.

```
1 public class Main {
2
3     static class InsufficientBalanceException extends Exception {
4         InsufficientBalanceException(String message) {
5             super(message);
6         }
7     }
8
9     static class Account {
10         int balance = 1000;
11
12         synchronized void deposit(int amount) {
13             balance = balance + amount;
14             System.out.println("Deposited: " + amount);
15         }
16
17         synchronized void withdraw(int amount)
18             throws InsufficientBalanceException {
19
20             if (amount > balance) {
21                 throw new InsufficientBalanceException("Insufficient balance");
22             }
23
24             balance = balance - amount;
25             System.out.println("Withdrawn: " + amount);
26         }
27     }
28
29     public static void main(String[] args) {
30
31         try {
32             Account account = new Account();
33             account.deposit(500);
34             account.withdraw(500);
35         } catch (InsufficientBalanceException e) {
36             System.out.println(e.getMessage());
37         }
38     }
39 }
```